

# PSimPy - Predictive and Probabilistic Simulation

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Want to build unified, robust and sustainable UQ workflows? <sup>[1]</sup>

PSimPy



## Highlights

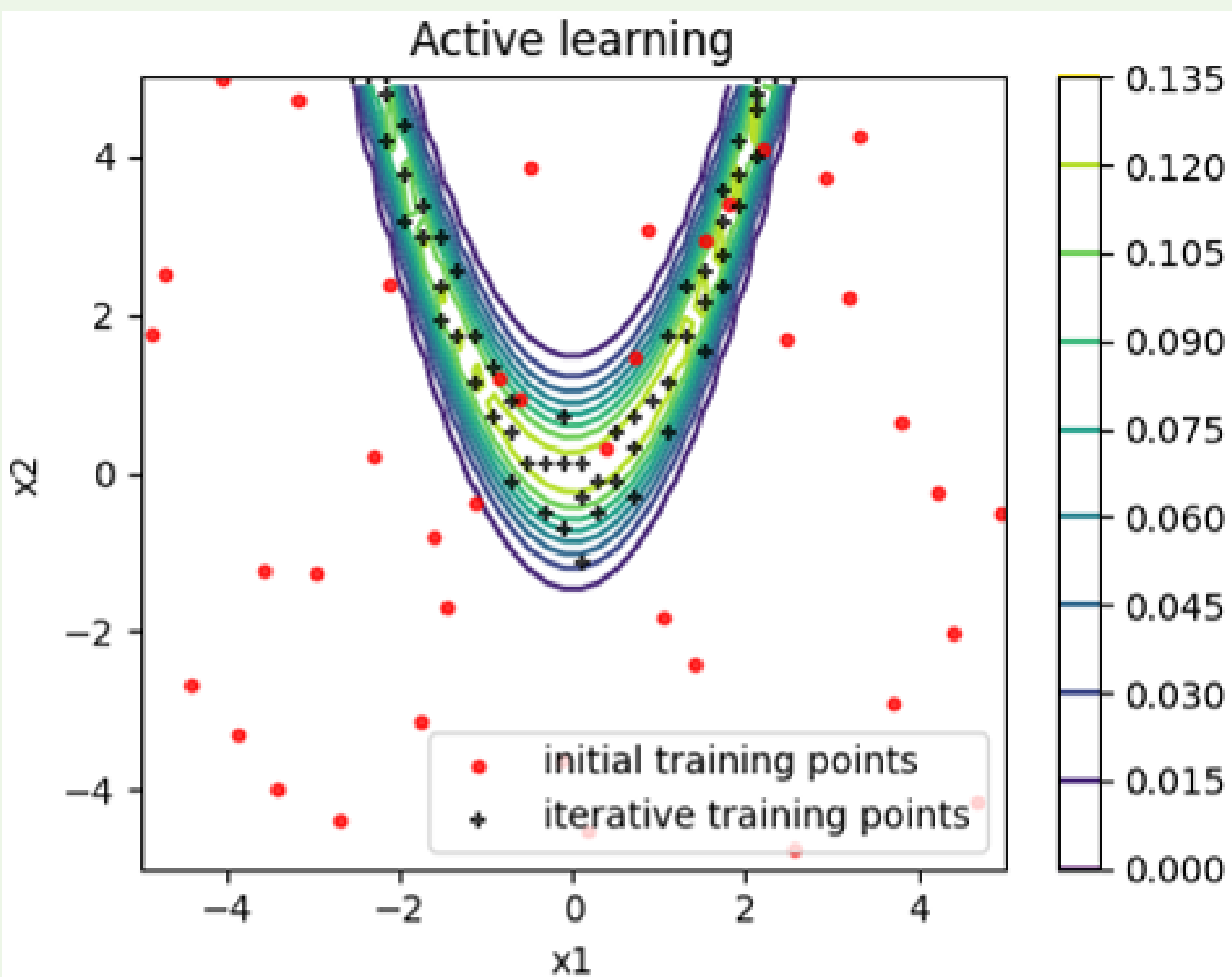
Robust GP emulator for your simulation.  
Save precious compute time!

|           | Training | Evaluation | 2500 Evals |
|-----------|----------|------------|------------|
| Simulator | -        | 32 min     | 56 days    |
| Emulator  | 4.5 days | 0.05 sec   | 2 min      |

- Gaussian process emulation
- Bayesian calibration
- Global sensitivity analysis
- Efficient sampling
- Active learning

## In Action

Quantify uncertainty, identify key drivers and calibrate with data.



Open source and Pythonic

[1] H. Zhao, A. Yildiz, N. Bagherinejad, and J. Kowalski, “PSimPy: GP emulation-based sensitivity analysis, uncertainty quantification and calibration of landslide simulators,” in *14th International Conference on Applications of Statistics and Probability in Civil Engineering, ICASP14*, Dublin, Ireland, 2023.